Solutions to Quiz 7

Question 1

Give an example of an irreducible polynomial of degree 4 in $\mathbb{F}_2[X]$.

Solution:

Given an irreducible polynomial P of degree 4, the field $\mathbb{F}_2[X]/\langle P\rangle$ is isomorphic to $\mathbb{F}_{16}$.

The only proper subfield of $\mathbb{F}_{16}$ is $\mathbb{F}_4$.

Now, the field $\mathbb{F}_{16}$ contains the 5-th roots of unity and these are *not* contained in $\mathbb{F}_4$. Thus, the polynomial

$$X^4 + X^3 + X^2 + X + 1$$

is irreducible in $\mathbb{F}_2[X]$.

Alternate Solution:

An irreducible polynomial of degree 1 or 2 in $\mathbb{F}_2[X]$ divides $X^4 - X$. This polynomial factors as $X(X+1)(X^2+X+1)$ in $\mathbb{F}_2[X]$.

Thus, if a polynomial of degree 4 is reducible:

- either it has a root in $\mathbb{F}_2$,
- or it is the square of $X^2 + X + 1$.

Note that $(X^2 + X + 1) = X^4 + X^2 + 1$. Secondly, if a polynomial has a root in $\mathbb{F}_2$, then does not have a constant term or it has an even number of terms. Thus, the irreducible polynomials of degree 4 in $\mathbb{F}_2[X]$ are:

$$X^4 + X^3 + X^2 + X + 1, X^4 + X^3 + 1, X^4 + X + 1$$

Question 2

What is the number of irreducible factors of the following polynomial in $\mathbb{F}_3[X]$ and what are their degrees?

$$\Phi_{13}(X) = \frac{X^{13} - 1}{X - 1} = \sum_{i=0}^{12} X^i$$

(Hint: You do not need to find the factors!)

Solution:

We note that the smallest n such that $3^n - 1$ is divisible by 13 is $n = 3$.

A root of $\Phi_{13}(X)$ is an element of order 13 in the multiplicative group.

Thus, *every* factor of $\Phi_{13}(X)$ is of degree 3. It follows that there are 4 such factors.